



**COMPETENCY STANDARD
FOR
PRODUCTION PROCESS IN FABRIC MANUFACTURING
(TEXTILE SECTOR)**

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance
Government of the People's Republic of Bangladesh

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The Competency Standard for **Production Process in Fabric Manufacturing** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry, so that individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades, which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The textile sector is the largest manufacturing and export-earning sector of Bangladesh, and fabric manufacturing is the critical mid-stream stage that links yarn production with wet processing and garment manufacturing. Both the woven and the knit segments of the industry depend on this stage for the dimensional accuracy, structural quality and defect-free surface of the fabric that is finally converted into apparel. Weaving preparatory operations, weaving on shuttle and shuttleless looms, circular knitting and flat bed knitting are labour-intensive, machine-paced operations in which operator skill directly determines machine efficiency, fabric quality, waste percentage and the cost competitiveness of the entire downstream chain. This Competency Standard has therefore been developed to prepare a competent entry-level workforce for the production process in fabric manufacturing.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and to ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competency and its corresponding Elements.

OVERVIEW:

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

1. provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
2. enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies.
3. encourages the development and delivery of flexible training that suits individual and industry requirements.
4. encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at a training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

1. Reference to Industry Sector, Occupational Title and Occupational Description
2. Unit code
3. Unit title
4. Unit descriptor
5. Unit of Competency
6. Elements and performance criteria
7. Variables and range statement
8. Evidence guides

Together all the parts of a Unit of Competency:

1. Describe a work activity
2. Guide the assessor in determining whether the candidate is competent.

This Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competency:

1. An overview of all Units of Competency for the occupation and their corresponding duration required for completion of training.
2. The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

DESCRIPTION OF THE OCCUPATION AND THE TRAINING COURSE:

Industry Sector	Textile
Occupation Title	Production Process in Fabric Manufacturing
Occupational Description	A worker competent in the Production Process in Fabric Manufacturing carries out the sequential operations required to convert yarn into woven and knitted fabric in a weaving or knitting factory. The occupation covers yarn winding, warping, sizing and loom preparatory operations (drawing-in, denting and knotting), weaving on shuttle and shuttleless looms, circular knitting and flat bed knitting, fabric inspection, grading and defect identification, together with fabric folding and packing, in-process quality checking, routine machine care and housekeeping. The worker performs these tasks under supervision, following standard operating procedures, machine setting sheets, fabric specifications, safety requirements and environmental practices of the factory, and records production and quality data using workplace formats.
Level of the Course	Basic Level Skills Training
Nominal Duration	360 Hours (03 Months)
Entry Qualification of Trainees	Grade VIII passed; age range 18-45
Target Number of Trainees	A total of 2,000 trainees to be trained by the end of 2028
Training Providers	Textile industries and training institutes having the capacity to deliver and run the training program in accordance with this Competency Standard and the Workshop/Lab Facility Standard.
Mode of Delivery	Institution-based and/or industry-based competency-based training combining theory, demonstration, guided practice and practical performance in an actual or simulated workplace.
Employment / Job Placement	The training providers (industries and training institutes) will arrange employment for at least 65% of the total enrollees upon completion of the training.
Assessment and Certification	Competency assessment will be conducted as per the Assessment Evidence Guide of each unit, in an assessment/training centre or in an actual or simulated workplace, by a nationally certified assessor or occupation-specific industry expert.
Possible Job Titles	Winding Operator, Warping Operator, Sizing Operator/Helper, Drawing-in and Denting Assistant, Loom Operator/Weaver, Rapier/Air-jet Loom Operator, Circular Knitting Machine Operator, Flat Bed Knitting Machine Operator, Fabric Inspector, Fabric Folding and Packing Assistant, Quality Checker (Trainee level).

UNITS & ELEMENTS AT A GLANCE:**Generic Competencies (20 Hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPFM-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-TEX-PPFM-02-G	Work in a Team Environment	1. Identify team goals and processes 2. Communicate and cooperate with team members 3. Work as a team member 4. Solve problems as a team member	10
Total Hours			20 Hrs.

Sector Specific Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPFM-01-S	Work in the Textile Sector	1. Identify the textile sector and its main activities 2. Understand basic roles in textile production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in textile work	10
SICIP-TEX-PPFM-02-S	Apply Green Practices in the Textile Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hours			20 Hrs.

Occupation Specific Competencies (320 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPFM-01-O	Interpret Yarns, Fabrics and Fabric Manufacturing Processes	1. Identify yarns and raw materials used in fabric manufacturing 2. Identify fabric types, structures and end uses 3. Identify machines, tools and equipment used in fabric manufacturing 4. Describe the fabric manufacturing process sequence 5. Interpret work instructions and production documents	50
SICIP-TEX-PPFM-02-O	Perform Winding and Warping Operations	1. Prepare for winding and warping operations 2. Carry out yarn winding operations 3. Carry out warping operations 4. Monitor package and beam quality and record production data	40
SICIP-TEX-PPFM-03-O	Perform Sizing and Loom Preparatory Operations	1. Prepare for sizing operations 2. Carry out the sizing operation 3. Perform drawing-in, denting and knotting operations 4. Check sized beam and warp quality and maintain records	40
SICIP-TEX-PPFM-04-O	Perform Weaving Operations	1. Prepare the loom for operation 2. Carry out the weaving operation 3. Attend warp and weft breakages and loom stoppages 4. Monitor fabric quality and control loom faults 5. Handle waste, cleaning and shift records	60
SICIP-TEX-PPFM-05-O	Perform Circular Knitting Operations	1. Prepare the circular knitting machine for operation 2. Carry out the circular knitting operation 3. Attend yarn breakage, needle faults and machine stoppages 4. Monitor knitted fabric quality and maintain records	60

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPFM-06-O	Perform Flat Bed Knitting Operations	1. Prepare the flat bed knitting machine for operation 2. Carry out flat bed knitting operations 3. Check panel quality, handle waste and maintain records	30
SICIP-TEX-PPFM-07-O	Perform Fabric Inspection, Grading and Defect Identification	1. Prepare for fabric inspection 2. Carry out fabric inspection 3. Identify fabric defects and determine their causes 4. Apply the fabric grading system 5. Report inspection results and take corrective action	40
Total Hours			320 Hrs.

Grand Total Duration of the Training Course	360 Hours (03 Months)
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COMPETENCY STANDARD: PRODUCTION PROCESS IN FABRIC MANUFACTURING

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPYM-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1. 1 <u>OHS policies</u> and safe operating procedures are read and understood. 1. 2 Safety signs and symbols are identified and followed. 1. 3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2. 1 OHS policies and procedures are followed and practiced. 2. 2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2. 3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to the authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4. 1 Alarms and warning devices are responded to. 4. 2 <u>Emergency response plans and procedures</u> are implemented. 4. 3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables:

Variable	Range (May include but not limited to)
1. OHS policies	1. 1 International OHS requirements

Variable	Range (May include but not limited to)
	1. 2 Bangladesh standards for OHS 1. 3 Bangladesh Labour Act and Rules 1. 4 Building Code 1. 5 Fire Safety Rules and Regulations 1. 6 Industry guidelines and mill safety manual
2. Personal Protective Equipment (PPE)	2. 1 Nose/face mask (dust mask) 2. 2 Ear plugs or ear muffs 2. 3 Hand gloves 2. 4 Safety shoes 2. 5 Head cap/helmet 2. 6 Apron and overalls 2. 7 Goggles and safety glasses 2. 8 Hair net
3. Hazards and risks	3.1 Physical hazards (moving machine parts, nip points, noise, vibration, heat) 3.2 Chemical hazards (lubricants, cleaning agents, solvents) 3.3 Biological hazards (bacteria, fungus in damp areas) 3.4 Ergonomic hazards (repetitive work, standing, manual lifting) 3.5 Electrical hazards 3.6 Fire hazards (cotton dust, fly accumulation, static electricity)
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid 4.5 Electrical accident response 4.6 Emergency machine stop procedures
5. First aid procedure	5. 1 Washing of open wound 5. 2 Washing chemically infected area 5. 3 Applying bandage 5. 4 Tourniquet 5. 5 Applying CPR (Cardiopulmonary Resuscitation) 5. 6 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures in a textile mill. 1.3 Emergency procedures. 1.3.1 Firefighting 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response.
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	1.4 Types of hazards (biological, chemical, physical, ergonomic and electrical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 Safety signs, symbols and colour codes. 1.8 OHS awareness and reporting requirements.
2. Underpinning Skills	2. 1 Identifying OHS policies and procedures 2. 2 Following personal work safety practices 2. 3 Selecting and using appropriate PPE 2. 4 Reporting hazards and risks 2. 5 Responding to emergency procedures 2. 6 Maintaining physical well-being in the workplace 2. 7 Performing basic firefighting using fire extinguishers 2. 8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 Safety signs and symbols 4.6 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Reported incidents arising from hazards and risks to the authority. 1.4 Implemented emergency response plans and procedures. 1.5 Applied basic first aid procedure.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration

	2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPYM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within the team and with other workers are identified.
2. Communicate and cooperate with team members	2. 1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2. 2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2. 3 Diversity in character is respected and valued in team functioning. 2. 4 Views and opinions of other team members are understood and valued. 2. 5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with the team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications, and workplace procedures. 3.3 Support to other team members is provided to ensure the team achieves its goals. 3.4 Agreed reporting lines are followed using standard operating procedures.
4. Solve problems as a team member	4.1 Current and potential problems faced by the team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables:

Variable	Range (May include but not limited to)
1. Forms of communication	1. 1 Agenda 1. 2 Simple reports such as progress and incident reports 1. 3 Job sheets and shift handover log 1. 4 Operational manuals and SOPs 1. 5 Brochures and promotional material 1. 6 Visual and graphic materials 1. 7 Standards 1. 8 OSH information 1. 9 Signs and symbols

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Team goals and collaborative decision-making processes 1. 2 Roles and responsibilities of team members 1. 3 Relationships within the team and with other workers 1. 4 Effective interpersonal skills to interact with team members 1. 5 Effective formal and informal forms of communication 1. 6 Value of diversity in team functioning 1. 7 Correct use of workplace terminology 1. 8 Team's duties, responsibilities, authorities, objectives and task requirements 1. 9 Support mechanisms for other members of the team to ensure achievement of goals 1. 10 Methods of identifying current and potential problems faced by a team 1. 11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2. 1 Identifying team goals and collaborative decision-making processes 2. 2 Identifying roles and responsibilities of team members 2. 3 Using effective interpersonal skills to interact with team members 2. 4 Using formal and informal forms of communication 2. 5 Understanding and valuing views and opinions of other team members 2. 6 Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures 2. 7 Supporting other members of the team to ensure the team achieves its goals 2. 8 Identifying current and potential problems faced by the team 2. 9 Identifying solutions to problems

	2. 10 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Workbooks

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified team goals and work processes. 1. 2 Communicated and cooperated with team members. 1. 3 Worked as a team member. 1. 4 Solved problems as a team member.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 4. Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: WORK IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPYM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the textile sector. It specifically includes the tasks of identifying the textile sector and its main activities, understanding basic roles in textile production processes, recognizing common tools and equipment used, and understanding simple quality requirements in textile work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the textile sector and its main activities	1. 1 The textile sector is identified within the context of national and global industries. 1. 2 <u>Main activities</u> of the textile sector are recognized. 1. 3 <u>Different stages of the textile value chain</u> are described in simple terms. 1. 4 Key workplaces in the textile sector are identified. 1. 5 Importance of the textile sector in the economy and the environment is understood.
2. Understand basic roles in textile production processes	2. 1 <u>Basic job roles in textile production</u> are identified and described. 2. 2 Simple production workflows and task sequences are understood. 2. 3 Teamwork and coordination in production processes are identified. 2. 4 Workplace safety roles and responsibilities are understood and followed. 2. 5 Shift working practices and reporting lines are understood.
3. Recognize common tools and equipment used	3.1 <u>Common tools used in textile processing</u> are identified correctly. 3.2 <u>Basic machines and equipment used in production</u> are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance and cleaning needs of tools and equipment are recognized.
4. Understand simple quality requirements in textile work	4.1 Basic quality standards in textile work are identified. 4.2 <u>Simple defects in textile products</u> are recognized. 4.3 Importance of maintaining quality during production is understood.

Elements of Competency	Performance Criteria
	4.4 Work output is checked against basic quality expectations. 4.5 Good workmanship practices are followed to ensure quality.

Range of Variables:

Variable	Range (May include but not limited to)
1. Main activities	1. 1 Fibre production and preparation 1. 2 Yarn manufacturing (spinning) 1. 3 Fabric manufacturing (weaving and knitting) 1. 4 Wet processing (dyeing, printing and finishing) 1. 5 Garment manufacturing 1. 6 Testing and quality assurance 1. 7 Packaging, storage and export
2. Different stages of the textile value chain	2. 1 Natural and man-made fibre production 2. 2 Fibre preparation and mixing 2. 3 Yarn manufacturing (spinning) 2. 4 Fabric manufacturing (weaving/knitting) 2. 5 Dyeing, printing and finishing 2. 6 Cutting, sewing and finishing of garments 2. 7 Packaging, distribution and export
3. Basic job roles in textile production	3.1 Blow room operator/helper 3.2 Card tenter 3.3 Draw frame and comber operator 3.4 Simplex (speed frame) operator 3.5 Ring frame sider and doffer 3.6 Winding and packing operator 3.7 Quality checker/laboratory assistant 3.8 Maintenance fitter/electrician 3.9 Line leader/supervisor
4. Common tools used in textile processing	4.1 Sliver and roving cutter, scissors and nippers 4.2 Hooks, brushes and dusting tools 4.3 Spanner, screwdriver and plier set 4.4 Oil can and grease gun 4.5 Feeler and setting gauges 4.6 Wrap reel, wrap block and electronic balance 4.7 Tachometer and hand lens 4.8 Vacuum cleaner and air blowing gun
5. Basic machines and equipment used in production	5. 1 Bale opener/plucker and blow room line 5. 2 Carding machine 5. 3 Draw frame (breaker and finisher) 5. 4 Lap former and comber 5. 5 Speed frame (simplex)

Variable	Range (May include but not limited to)
	5. 6 Ring frame 5. 7 Winding machine/autoconer 5. 8 Humidification and waste collection plant 5. 9 Testing laboratory equipment
6. Simple defects in textile products	6. 1 Neps and slubs 6. 2 Thick and thin places 6. 3 Count variation and uneven yarn 6. 4 High hairiness 6. 5 Foreign fibre and contamination 6. 6 Sliver and roving breakage 6. 7 Soft/hard and damaged packages 6. 8 Oil and dirt stains

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Structure and importance of the textile sector in the national and global economy. 1. 2 Main activities and stages of the textile value chain. 1. 3 Position of yarn manufacturing within the textile value chain. 1. 4 Basic job roles, responsibilities and reporting lines in textile production. 1. 5 Simple production workflows, task sequences and shift practices. 1. 6 Common tools, machines and equipment used in textile production and their functions. 1. 7 Basic maintenance and cleaning needs of tools and equipment. 1. 8 Basic quality standards and common defects in textile products. 1. 9 Importance of quality, productivity and good workmanship in the workplace.
2. Underpinning Skills	2. 1 Identifying the textile sector and its main activities 2. 2 Describing the stages of the textile value chain in simple terms 2. 3 Understanding basic job roles and responsibilities in textile production processes 2. 4 Following simple production workflows, task sequences and teamwork practices 2. 5 Applying workplace safety roles and responsibilities in production activities 2. 6 Recognizing common tools, machines and equipment and their functions 2. 7 Recognizing basic maintenance and cleaning needs of tools and equipment

	2. 8 Recognizing simple defects and checking work output against basic quality requirements 2. 9 Following good workmanship practices to ensure consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Charts, models and samples of fibre, sliver, roving and yarn 4.3 Multimedia projector, laptop and audio-visual clips 4.4 Learning manual and handouts 4.5 Stationery

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified the textile sector and its main activities. 1. 2 Understood basic roles in textile production processes. 1. 3 Recognized common tools and equipment used. 1. 4 Understood simple quality requirements in textile work.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPYM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in the textile sector. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1. 1 Basic concepts of environmental sustainability and green practices are explained clearly. 1. 2 <u>Environmental impacts of textile production</u> processes are identified. 1. 3 <u>Resource efficiency concepts</u> are interpreted. 1. 4 Waste reduction, reuse and recycling principles are described in relation to yarn manufacturing operations. 1. 5 <u>Eco-friendly practices in the textile sector</u> are recognized and related to workplace activities.
2. Minimize resource use in the workplace	2. 1 Electricity, water, compressed air and raw materials are used efficiently according to workplace requirements. 2. 2 <u>Unnecessary consumption of resources</u> is identified and avoided during operations. 2. 3 Machines and equipment are operated in energy-efficient modes where applicable. 2. 4 Material wastage is minimized through careful handling and accurate measurement. 2. 5 <u>Resource-saving practices</u> are applied where possible. 2. 6 Leakage, spillage and wastage of materials are promptly controlled and reported. 2. 7 <u>Workplace practices</u> for resource conservation are followed.
3. Implement waste management practices	3.1 <u>Types of waste</u> generated in yarn manufacturing operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 <u>Reusable and recyclable materials</u> are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or areas. 3.5 Waste disposal is carried out following environmental rules and organizational procedures.

Elements of Competency	Performance Criteria
	3.6 Hazardous waste is handled carefully according to safety guidelines. 3.7 7. Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables:

Variable	Range (May include but not limited to)
1. Environmental impacts of textile production	1. 1 High electrical energy consumption of spinning machinery and humidification plants 1. 2 Cotton dust, fibre fly and micro-fibre emission 1. 3 Noise pollution from machines and compressors 1. 4 Solid waste generation (droppings, comber noil, soft and hard waste) 1. 5 Water consumption and effluent from wet processing 1. 6 Air emission and greenhouse gases from boilers and generators 1. 7 Used oil, grease and contaminated cotton waste
2. Resource efficiency concepts	2. 1 Energy-efficient motors and inverter/variable frequency drives 2. 2 Control of compressed air leakage and optimized air pressure 2. 3 Optimized humidification and ventilation control 2. 4 Reduction of soft and hard waste percentage 2. 5 Preventive maintenance to reduce power consumption 2. 6 Energy-efficient lighting and switching-off practices 2. 7 Optimum machine loading and reduction of idle running
3. Eco-friendly practices in the textile sector	3.1 Recycling of soft waste back into mixing where permitted 3.2 Use of recycled, organic or sustainably sourced fibre 3.3 Dust collection, filter and waste recovery systems 3.4 Compliance with Department of Environment (DoE) rules and regulations 3.5 Effluent treatment plant (ETP) and rainwater harvesting 3.6 Reuse of cans, bobbins, cones, cartons and packaging materials 3.7 Green building and energy audit practices
4. Unnecessary consumption of resources	4.1 Idle running of machines and motors 4.2 Compressed air leakage in pipelines and guns 4.3 Excess lighting in unoccupied areas 4.4 Over-cleaning with air guns instead of vacuum 4.5 Excess lubricant and cleaning material use 4.6 Excessive good fibre loss with droppings and noil

Variable	Range (May include but not limited to)
5. Resource-saving practices	5. 1 Switching off machines, lights and fans when not required 5. 2 Reporting air, water and oil leakage promptly 5. 3 Using vacuum cleaning instead of air blowing where possible 5. 4 Reusing cans, tubes, cones and packaging materials 5. 5 Maintaining correct machine settings to reduce waste 5. 6 Recording and monitoring waste and energy use
6. Workplace practices	6. 1 Standard operating procedures (SOPs) 6. 2 Environmental policy of the organization 6. 3 Housekeeping and 5S practices 6. 4 Energy and waste monitoring records 6. 5 Instructions from supervisors and safety notices
7. Types of waste	7.1 Blow room droppings and card droppings/flat strips 7.2 Comber noil 7.3 Sliver, roving and pneumafil soft waste 7.4 Hard waste from winding (yarn and thread waste) 7.5 Damaged cans, bobbins, cones and tubes 7.6 Packaging waste (poly bags, bale wrappers, straps, cartons) 7.7 Used oil, grease and oil-contaminated cotton (hazardous waste) 7.8 Used filters, sweeps and floor waste
8. Reusable and recyclable materials	8.1 Usable soft waste for back-mixing 8.2 Comber noil for coarse count and open-end spinning 8.3 Cans, bobbins, tubes and cones 8.4 Cartons, poly bags and bale straps 8.5 Metal and plastic scrap 8.6 Paper and stationery waste

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Basic concepts of environmental sustainability, green practices and climate-resilient production. 1. 2 Environmental impacts of textile and yarn manufacturing operations. 1. 3 Resource efficiency concepts relating to energy, water, compressed air and raw material. 1. 4 Principles of waste reduction, reuse and recycling in a spinning mill. 1. 5 Eco-friendly practices and technologies applied in the textile sector. 1. 6 Types of waste generated at each stage of yarn manufacturing and permissible waste levels.
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	1. 7 Waste segregation, storage, disposal methods and compliance with environmental regulations. 1. 8 Hazardous waste handling and workplace environmental safety procedures. 1. 9 Relevant national environmental rules and organizational environmental policy.
2. Underpinning Skills	2. 1 Interpreting green concepts and relating them to workplace activities 2. 2 Identifying environmental impacts of yarn manufacturing operations 2. 3 Applying resource efficiency principles in workplace operations 2. 4 Using electricity, compressed air, water and raw materials efficiently 2. 5 Identifying and avoiding unnecessary consumption of resources 2. 6 Operating machines and equipment in energy-efficient modes where applicable 2. 7 Handling materials carefully to reduce wastage and improve accuracy 2. 8 Controlling and reporting leakage, spillage and resource wastage promptly 2. 9 Identifying and segregating different types of waste generated in yarn manufacturing 2. 10 Separating reusable and recyclable materials from non-recyclable waste 2. 11 Handling hazardous waste according to safety guidelines and regulations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Standard operating procedures 4. 3 Workplace documents, signs and symbols 4. 4 Codes of conduct and environmental policy 4. 5 Waste segregation bins and storage area 4. 6 Tools, equipment and facilities appropriate to the process or activities 4. 7 Materials relevant to the proposed activity

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Interpreted green concepts. 1. 2 Minimized resource use in the workplace. 1. 3 Implemented waste management practices.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: INTERPRET YARNS, FABRICS AND FABRIC MANUFACTURING PROCESSES	Nominal Duration: 50 Hrs.	Unit Code: SICIP-TEX-PPFM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret yarns, fabrics and fabric manufacturing processes. It specifically includes the tasks of identifying yarns and raw materials used in fabric manufacturing, identifying fabric types, structures and end uses, identifying machines, tools and equipment used in fabric manufacturing, describing the fabric manufacturing process sequence, and interpreting work instructions and production documents.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify yarns and raw materials used in fabric manufacturing	1.1 <u>Types of yarn</u> used for weaving and knitting are identified according to fibre, structure and spinning system. 1.2 <u>Yarn count systems</u> are interpreted and simple conversions are performed. 1.3 Yarn quality parameters affecting weaving and knitting performance are recognized. 1.4 <u>Yarn package forms and identification documents</u> are interpreted.
2. Identify fabric types, structures and end uses	2.1 <u>Types of fabric</u> are identified according to the method of manufacture. 2.2 Basic woven structures are identified and described. 2.3 Basic knitted structures are identified and described.
3. Identify machines, tools and equipment used in fabric manufacturing	3.1 <u>Machines used in fabric manufacturing</u> are identified. 3.2 <u>Hand tools and measuring instruments</u> are identified and their functions are described. 3.3 Safety requirements for using machines, tools and equipment are identified.
4. Describe the fabric manufacturing process sequence	4.1 <u>Stages of the weaving process</u> are described in the correct sequence. 4.2 <u>Stages of the knitting process</u> are described in the correct sequence. 4.3 Differences between woven, weft knitted and warp knitted fabric manufacturing routes are explained.
5. Interpret work instructions and production documents	5.1 Production documents are identified and interpreted. 5.2 Work orders, fabric specification sheets and machine setting sheets are read and understood.

Elements of Competency	Performance Criteria
	5.3 Standard operating procedures (SOPs) are followed as instructed. 5.4 Workplace terminology is used correctly during communication. 5.5 Information is recorded and reported using workplace formats.

Range of Variables:

Variable	Range (May include but not limited to)
1. Types of yarn	1.1 Carded and combed yarn 1.2 Blended yarn (cotton/polyester, cotton/viscose, cotton/lycra) 1.3 Ring spun, 1.4 rotor/open-end 1.5 Single and ply (doubled) yarn 1.6 Filament yarn (POY, FDY, DTY) 1.7 Core-spun and elastane (spandex) covered yarn 1.8 Slub and fancy yarn
2. Yarn count systems	2.1 Indirect systems 2.1.1 English count (Ne) 2.2 Direct systems 2.2.1 Tex 2.2.2 Denier 2.3 Twist per inch (TPI) 2.4 Twist direction (S and Z)
3. Yarn package forms and identification documents	3.1 Cone, cheese, ring cop and pirn 3.2 Hank and warper's/weaver's beam
4. Types of fabric	4.1 Woven fabric 4.2 Weft knitted fabric (circular and flat bed) 4.3 Warp knitted fabric 4.4 Non-woven fabric 4.5 Braided and narrow fabric
5. Machines used in fabric manufacturing	5.1 Cone/cheese winding machine and pirn winding machine 5.2 Direct (beam) warping machine and sectional warping machine 5.3 Sizing machine with size cooking unit 5.4 Drawing-in frame and denting arrangement 5.5 Warp tying/knotting machine 5.6 Shuttle loom (power loom) 5.7 Shuttleless looms (rapier, air-jet, water-jet, projectile) 5.8 Single jersey, rib and interlock circular knitting machines 5.9 Fabric inspection machine

Variable	Range (May include but not limited to)
6. Hand tools and measuring instruments	6.1 Reed hook, drawing hook and denting hook 6.2 Scissors, nippers and thread cutter 6.3 Oil can and grease gun 6.4 Pick glass/counting glass and magnifier 6.5 GSM cutter and electronic balance
7. Stages of the weaving process	7.1 Yarn receiving, checking and conditioning 7.2 Cone/cheese winding 7.3 Warping (direct or sectional) 7.4 Sizing 7.5 Drawing-in, denting and knotting/tying-in 7.6 Weft preparation (pirn winding, where applicable) 7.7 Weaving on the loom 7.8 Fabric inspection and grading 7.9 Mending and spot cleaning 7.10 Folding, rolling, packing and dispatch
8. Stages of the knitting process	8.1 Yarn receiving, checking and conditioning 8.2 Yarn winding and waxing (where required) 8.3 Machine setting, needle and sinker checking 8.4 Creeling and yarn feeding 8.5 Sample knitting and GSM/stitch length approval 8.6 Bulk knitting on circular or flat bed machine 8.7 Fabric roll doffing or panel separation 8.8 Fabric inspection and grading 8.9 Weighing, folding and packing

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Types of yarn used in weaving and knitting, their manufacturing systems and characteristics. 1.2 Yarn count systems, count conversion, twist, twist direction and twist multiplier. 1.3 Yarn quality parameters and their influence on weaving and knitting performance. 1.4 Classification of fabrics by method of manufacture. 1.5 Basic woven structures, their construction, appearance and identification. 1.6 Basic knitted structures, their construction, appearance and identification. 1.7 Machines used at each stage of fabric manufacturing and their functions. 1.8 Hand tools and measuring instruments used in the weaving and knitting departments. 1.9 Sequence and objectives of the weaving and knitting process.
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	1.10 Process parameters and their influence on fabric quality and productivity. 1.11 Production documents, machine setting sheets, SOPs and workplace terminology.
2. Underpinning Skills	2.1 Identifying yarn types by appearance and simple physical examination 2.2 Determining yarn count using wrap reel and electronic balance 2.3 Identifying woven and knitted fabric structures using a pick glass 2.4 Interpreting fabric specification sheets and design sheets 2.5 Determining face and back sides and direction of fabric 2.6 Identifying machines, tools and measuring instruments used in fabric manufacturing 2.7 Describing the weaving and knitting process sequence and the objective of each stage 2.8 Identifying in-process material forms and tracing material flow 2.9 Interpreting work orders, machine setting sheets and production documents 2.10 Recording and reporting information using workplace formats
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Yarn sample board with yarns of different counts, types and packages 4.3 Fabric sample board with woven and knitted structures 4.4 Charts, models and flow diagrams of the weaving and knitting processes 4.5 Wrap reel, wrap block and electronic balance 4.6 GSM cutter, GSM balance and pick glass 4.7 Hand tools and measuring instruments 4.8 Multimedia projector, laptop and audio-visual clips 4.9 Learning manual, SOPs and sample production formats

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified yarns and raw materials used in fabric manufacturing. 1.2 Identified fabric types, structures and end uses. 1.3 Identified machines, tools and equipment used in fabric manufacturing. 1.4 Described the fabric manufacturing process sequence. 1.5 Interpreted work instructions and production documents.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM WINDING AND WARPING OPERATIONS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-TEX-PPFM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform winding and warping operations. It specifically includes the tasks of preparing for winding and warping operations, carrying out yarn winding operations, carrying out warping operations, and monitoring package and beam quality and recording production data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for winding and warping operations	1.1 Work order, yarn details and lot information are obtained and understood. 1.2 Machines, creel and work area are checked for cleanliness and readiness. 1.3 <u>Safety devices and guards</u> are checked before start-up. 1.4 Required materials and accessories are arranged at the workstation. 1.5 Abnormal conditions found during checking are reported to the authority.
2. Carry out yarn winding operations	2.1 Yarn packages are creeled and threaded through the <u>yarn path</u> correctly. 2.2 Winding parameters are set according to yarn type and package requirement. 2.3 Yarn clearer settings are checked and applied as instructed. 2.4 <u>Winding faults</u> are identified and corrected during operation. 2.5 Package doffing is carried out and packages are placed in designated trolleys. 2.6 Yarn waste is collected and segregated as per workplace procedure.
3. Carry out warping operations	3.1 <u>Warping method</u> is applied according to the work order. 3.2 Creel is loaded and yarn ends are drawn through tensioners, guides and stop motions. 3.3 Warping parameters are set as per the warp plan. 3.4 Number of ends, beam width and warp length are maintained according to specification. 3.5 Colour and end sequence are maintained according to the warp plan. 3.6 Broken ends are attended and joined promptly using the standard knotting method. 3.7 <u>Warping faults</u> are identified and corrected or reported.

Elements of Competency	Performance Criteria
	3.8 Beam doffing, leasing, marking and identification are completed correctly.
4. Monitor package and beam quality and record production data	4.1 Package and beam quality parameters are checked at specified intervals. 4.2 Yarn tension and beam density are verified against the standard. 4.3 <u>Common faults</u> are identified and reported to the authority. 4.4 Waste percentage is checked against the standard. 4.5 Production and quality data are recorded in the <u>prescribed format</u> . 4.6 Machine, creel and work area are cleaned after completion of work.

Range of Variables:

Variable	Range (May include but not limited to)
1. Safety devices and guards	1.1 Emergency stop buttons and pull cords 1.2 Machine guards, covers and interlock switches 1.3 Creel guards and yarn stop motions 1.4 Beam flange guard and doffing arrangement 1.5 Earthing and electrical safety devices 1.6 Warning signs and floor markings 1.7 Lockout-tagout arrangement
2. Yarn path	2.1 Package holder and unwinding accelerator/balloon breaker 2.2 Yarn guide and ceramic eyelet 2.3 Tensioner (disc, gate or capstan type) 2.4 Yarn clearer/slub catcher (mechanical or electronic) 2.5 Yarn detector and stop motion 2.6 Traverse drum and yarn guide finger
3. Winding faults	3.1 Soft and hard package 3.2 Ribbon formation and patterning 3.3 Stitching and sloughing-off 3.4 Bad or weak splice/knot 3.5 Uneven package shape, bulged nose or base 3.6 Overfilled and underfilled package 3.7 Mixed count, mixed lot and shade mix 3.8 Oil, dirt and grease stain on package 3.9 Broken and entangled yarn on the package surface
4. Warping method	4.1 Direct (beam) warping for single-colour and large-lot production 4.2 Sectional (drum) warping for multi-colour and small-lot production

Variable	Range (May include but not limited to)
5. Warping faults	5.1 Missing, broken and lapped ends 5.2 Crossed ends and wrong end sequence 5.3 Wrong colour pattern and repeat error 5.4 Soft, hard or unevenly built beam 5.5 Entangled and snarled ends
6. Common faults	6.1 Soft or hard package and beam 6.2 Stitched or slough-off package 6.3 Ribbon formation 6.4 Bad splice and slipping knot 6.5 Missing, crossed and lapped ends 6.6 Uneven beam build and ridge formation 6.7 Oil, dirt and rust stains 6.8 Mixed count, mixed lot and shade variation 6.9 Excessive waste generation
7. Prescribed format	7.1 Winding production log sheet 7.2 Warping production log sheet 7.3 Beam and package card 7.4 Clearer cut and splice report 7.5 Machine stoppage and breakdown register 7.6 Quality checking sheet 7.7 Waste record 7.8 Shift handover log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Types of winding machines, package forms and their applications. 1.2 Yarn path in a winding machine and the function of each element. 1.3 Winding parameters and their effect on package quality and unwinding performance. 1.4 Splicing and knotting methods, splice quality requirements and testing. 1.5 Waxing of knitting yarn and its purpose. 1.6 Winding faults, their causes, effects and remedies. 1.7 Creel types, creeling methods and yarn tensioning arrangements. 1.8 Warping parameters, warp plan and colour order interpretation. 1.9 Warping faults, their causes, effects and remedies. 1.10 Package and beam quality parameters and methods of checking. 1.11 Waste generation in winding and warping and permissible waste limits.
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2. Underpinning Skills	2.1 Checking machines, creel and safety devices before start-up 2.2 Creeling yarn packages and threading the yarn path correctly 2.3 Setting winding parameters according to yarn type and package requirement 2.4 Performing splicing and knotting within the specified standard 2.5 Identifying and correcting winding faults during operation 2.6 Setting warping parameters as per the warp plan 2.7 Maintaining number of ends, beam width, warp length and colour sequence 2.8 Identifying and correcting warping faults 2.9 Doffing, leasing, marking and identifying beams 2.10 Checking package and beam quality parameters 2.11 Measuring yarn tension and beam density 2.12 Collecting, segregating and recording waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Cone/cheese winding machine and pirn winding machine 4.3 Direct (beam) warping machine and sectional warping machine 4.4 Creel with tensioners, guides and stop motions 4.5 Yarn packages of different counts and types 4.6 Splicer, knotter and knotting tools 4.7 Yarn tension meter and electronic balance 4.8 Wrap reel and wrap block 4.9 Trolleys, pallets and beam-handling arrangement 4.10 Learning manual, SOPs and sample production formats 4.11 Multimedia projector, laptop and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared for winding and warping operations. 1.2 Carried out yarn winding operations. 1.3 Carried out warping operations.
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	1.4 Monitored package and beam quality and recorded production data.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM SIZING AND LOOM PREPARATORY OPERATIONS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-TEX-PPFM-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform sizing and loom preparatory operations. It specifically includes the tasks of preparing for sizing operations, carrying out the sizing operation, performing drawing-in, denting and knotting operations, and checking sized beam and warp quality and maintaining records.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for sizing operations	1.1 Work order, warp plan and <u>size recipe</u> are obtained and understood. 1.2 Sizing machine, size box, creel and headstock are checked for cleanliness and readiness. 1.3 Sizing materials and chemicals are collected, weighed and checked for suitability. 1.4 <u>Safety devices and guards</u> are checked before start-up.
2. Carry out the sizing operation	2.1 Size paste is prepared in the <u>cooking vessel</u> following the recipe and the correct mixing sequence. 2.2 <u>Size paste parameters</u> are checked and maintained within the specified limit. 2.3 Warper's beams are creeled and the warp sheet is passed through the size box, drying zone and headstock. 2.4 <u>Sizing process parameters</u> are set and maintained as per the work order. 2.5 Size pick-up percentage, moisture content and stretch are maintained within the standard. 2.6 Splitting of the warp sheet and lease rod arrangement are maintained correctly. 2.7 Broken ends during sizing are attended and joined without disturbing the sheet. 2.8 Sized beam is doffed, marked and identified correctly.
3. Perform drawing-in, denting and knotting operations	3.1 <u>Drawing-in accessories</u> are selected according to the fabric design. 3.2 <u>Drafting and denting plan</u> is interpreted from the design sheet. 3.3 Warp ends are drawn through drop wires, healds and reed in the correct order. 3.4 Denting is completed maintaining the specified ends per dent and selvedge arrangement. 3.5 Knotting/tying-in is performed when the same design is repeated.

Elements of Competency	Performance Criteria
4. Check sized beam and warp quality and maintain records	4.1 Sized beam quality parameters are checked at specified intervals. 4.2 Size pick-up, moisture content and elongation are measured and recorded. 4.3 <u>Common sizing faults</u> are identified and reported to the authority. 4.4 Waste percentage is checked against the standard. 4.5 Production and quality data are recorded in the <u>prescribed format</u>. 4.6 Machine, size box, cooking vessel and work area are cleaned after completion of work.

Range of Variables:

Variable	Range (May include but not limited to)
1. Size recipe	1.1 Adhesive (native starch, modified starch, PVA, CMC, acrylic binder) 1.2 Softener and lubricant (wax, tallow, mineral oil) 1.3 Water quantity and total solid content 1.4 Separate recipes for cotton, polyester/cotton and filament warp
2. Safety devices and guards	2.1 Emergency stop buttons and pull cords 2.2 Guards over drying cylinders, squeeze rollers and headstock 2.3 Steam valve, pressure gauge and safety relief valve 2.4 Temperature and level alarms 2.5 Warning signs, floor markings and anti-slip flooring 2.6 Lockout-tagout arrangement
3. Cooking vessel	3.1 High-pressure cooker 3.2 Size mixing tank with stirrer 3.3 Size service/supply tank with jacket heating 3.4 Size box with immersion and squeeze rollers 3.5 Refractive index
4. Size paste parameters	4.1 Viscosity 4.2 Concentration and total solid content (Brix/refractive index) 4.3 Temperature of the paste in the cooker and size box 4.4 pH value 4.5 Cooking time and stirring speed 4.6 Level of size paste in the size box
5. Sizing process parameters	5.1 Machine (sizing) speed 5.2 Size box temperature 5.3 Squeeze roller pressure and number of nips 5.4 Immersion roller position and number of dips

Variable	Range (May include but not limited to)
	5.5 Drying cylinder temperature and steam pressure 5.6 Size pick-up percentage 5.7 Beam pressure, beam density and winding tension 5.8 Splitting arrangement and lease rod setting
6. Drawing-in accessories	6.1 Drop wires (drop pins) - closed and open end 6.2 Heald wires, healds and heald frames/shafts 6.3 Reed of specified count and width 6.4 Lease rods and lease bands 6.5 Beam gaiting accessories and beam gears 6.6 Reed hook, drawing hook and denting hook 6.7 Heald frame connecting rods and springs 6.8 Number of heald shafts according to the design repeat
7. Drafting and denting plan	7.1 Straight draft 7.2 Pointed and V draft 7.3 Skip and broken draft
8. Common sizing faults	8.1 Over-sized (hard) and under-sized (soft) warp 8.2 Sticky ends and improper splitting 8.3 Uneven size pick-up across the sheet 8.4 Size lumps, shell and skin formation 8.5 Uneven moisture content and over-drying 8.6 Broken, crossed and lapped ends 8.7 Soft, ridged or unevenly built beam 8.8 Uneven beam edges and yarn falling over the flange
9. Prescribed format	9.1 Sizing production log sheet 9.2 Size recipe and cooking record 9.3 Size pick-up, moisture and stretch record 9.4 Beam card and set card 9.5 Drawing-in, denting and knotting job card 9.6 Machine stoppage and breakdown register 9.7 Waste record 9.8 Shift handover log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Objectives and importance of sizing in the weaving preparatory process. 1.2 Construction and zones of a sizing machine (creel, size box, drying, headstock). 1.3 Sizing process parameters and their effect on warp quality and weaving performance. 1.4 Size pick-up, moisture content and stretch: definition, measurement and control. 1.5 Sizing faults, their causes, effects and remedies.
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	1.6 Purpose and methods of drawing-in, denting, knotting and gaiting. 1.7 Drop wires, healds, heald frames and reeds: types, specification and selection. 1.8 Drafting plans, lifting plans and denting plans and their interpretation. 1.9 Hand knotting and machine tying-in methods. 1.10 Waste generation in sizing and permissible waste limits. 1.11 Safe handling and storage of sizing chemicals, steam and hot surfaces.
2. Underpinning Skills	2.1 Preparing size paste following the recipe and mixing sequence 2.2 Measuring and controlling viscosity, concentration, temperature and pH of size paste 2.3 Creeling warper's beams and passing the warp sheet through the machine 2.4 Setting and maintaining sizing process parameters 2.5 Measuring size pick-up, moisture content and stretch 2.6 Selecting drawing-in accessories according to the design 2.7 Drawing warp ends through drop wires, healds and reed 2.8 Denting with the specified ends per dent and selvedge arrangement 2.9 Performing hand knotting and operating the warp tying machine 2.10 Checking beam assembly for missing, crossed and double ends 2.11 Identifying sizing faults and their causes
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Sizing machine with size box, drying cylinders and headstock 4.3 Sizing chemicals and materials 4.4 Viscometer, refractometer/Brix meter, pH meter and thermometer 4.5 Moisture meter and electronic balance 4.6 Warper's beams and weaver's beams

	4.7 Drawing-in frame with drop wires, healds, heald frames and reeds 4.8 Reed hook, drawing hook and denting hook 4.9 Warp tying/knotting machine 4.10 Waste segregation bins 4.11 Learning manual, SOPs, MSDS and sample production formats 4.12 Multimedia projector, laptop and audio-visual clips
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared for sizing operations. 1.2 Carried out the sizing operation. 1.3 Performed drawing-in, denting and knotting operations. 1.4 Checked sized beam and warp quality and maintained records.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM WEAVING OPERATIONS	Nominal Duration: 60 Hrs.	Unit Code: SICIP-TEX-PPFM-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform weaving operations. It specifically includes the tasks of preparing the loom for operation, carrying out the weaving operation, attending warp and weft breakages and loom stoppages, monitoring fabric quality and controlling loom faults, and handling waste, cleaning and shift records.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the loom for operation	1.1 Work order, <u>fabric specification</u> and machine setting sheet are obtained and understood. 1.2 Safety devices and guards are checked before start-up. 1.3 Sized beam, healds, reed, drop wires and weft package are gaited and mounted correctly. 1.4 Loom settings are verified against the machine setting sheet.
2. Carry out the weaving operation	2.1 <u>Primary, secondary and auxiliary motions</u> are observed for correct functioning. 2.2 Warp tension, pick density and fabric take-up are maintained as per specification. 2.3 Weft insertion and selvedge formation are monitored continuously. 2.4 Colour and weft pattern sequence are maintained according to the design.
3. Attend warp and weft breakages and loom stoppages	3.1 Warp breakages are attended and ends are joined using the standard knot without disturbing the weave. 3.2 Weft breakage and weft exhaustion are attended and weft is replenished promptly. 3.3 Loom stop motions are attended and reset correctly. 3.4 Starting marks, thin places and thick places are prevented while restarting the loom. 3.5 Faults beyond the operator's scope are reported to the supervisor or mechanic.
4. Monitor fabric quality and control loom faults	4.1 Woven fabric is checked on the loom at specified intervals. 4.2 <u>Common woven fabric defects</u> are identified and their causes are determined. 4.3 Corrective actions within the operator's scope are taken to remove defects. 4.4 Fabric width, pick density and selvedge quality are verified against specification. 4.5 Defective fabric portions are marked and reported.

Elements of Competency	Performance Criteria
5. Handle waste, cleaning and shift records	5.1 Warp, weft and fabric waste are collected and segregated as per workplace procedure. 5.2 Waste percentage is maintained within the permissible limit. 5.3 Loom, under-loom area and accessories are cleaned as per the cleaning schedule. 5.4 Routine lubrication points are attended as per the schedule. 5.5 Production, efficiency and quality data are recorded in the <u>prescribed format</u>. 5.6 Shift handover is completed with correct and complete information.

Range of Variables:

Variable	Range (May include but not limited to)
1. Fabric specification	1.1 Warp count and weft count 1.2 Ends per inch (EPI) and picks per inch (PPI) 1.3 Reed count, reed width and ends per dent 1.4 GSM 1.5 Warp and weft crimp and waste percentage 1.6 Selvage type and selvage width 1.7 Fabric length per roll and total order quantity
2. Primary, secondary and auxiliary motions	2.1 Primary motions 2.1.1 Shedding 2.1.2 Picking (weft insertion) 2.1.3 Beat-up 2.2 Secondary motions 2.2.1 Take-up motion 2.2.2 Let-off motion 2.3 Auxiliary (tertiary) motions 2.3.1 Warp stop motion 2.3.2 Weft stop motion
3. Common woven fabric defects	3.1 Broken and missing ends and picks 3.2 Double end and double pick 3.3 Starting mark and stop mark 3.4 Reed mark and reed cut 3.5 Temple mark and temple cut 3.6 Float and wrong lifting 3.7 Weft bar and shade bar 3.8 Oil stain, dirt mark and rust mark 3.9 Loose, tight and cut selvage 3.10 Hole, cut and tear 3.11 Snarl, slub and knot 3.12 Misspick and weft crack 3.13 Wrong draw and wrong denting

Variable	Range (May include but not limited to)
	3.14 Uneven fabric width and uneven pick density
4. Prescribed format	4.1 Loom production and efficiency report 4.2 Pick counter reading sheet 4.3 Loom stoppage and breakdown register 4.4 Warp and weft breakage record 4.5 Fabric roll card and beam card 4.6 On-loom inspection sheet 4.7 Waste record 4.8 Lubrication and cleaning checklist 4.9 Shift handover log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Classification of looms and the working principle of shuttle and shuttleless looms. 1.2 Comparative features of rapier, air-jet, water-jet and projectile weaving. 1.3 Loom settings and their effect on fabric quality and machine performance. 1.4 Primary, secondary and auxiliary motions of the loom and their timing relationship. 1.5 Shedding devices: tappet, dobby and jacquard, and their selection for different designs. 1.6 Loom stop motions, their function and correct resetting. 1.7 Common woven fabric defects, their causes and remedies. 1.8 Waste generation in weaving and permissible waste limits. 1.9 Cleaning and lubrication schedules for looms. 1.10 Safe working procedures, machine guarding and lockout-tagout in the weaving shed. 1.11 Recording and reporting production, stoppage and quality data.
2. Underpinning Skills	2.1 Checking the loom, safety devices and guards before start-up 2.2 Mounting and threading the weft package and weft accumulator 2.3 Starting and stopping the loom following SOPs 2.4 Observing the primary, secondary and auxiliary motions during running 2.5 Operating tappet, dobby and jacquard shedding as per the design 2.6 Monitoring weft insertion and selvedge formation 2.7 Maintaining colour and weft pattern sequence 2.8 Attending warp breakages and joining ends with the standard knot

	2.9 Attending weft breakage and replenishing weft promptly 2.10 Identifying woven fabric defects on the loom and determining their causes 2.11 Doffing, marking and handling fabric rolls 2.12 Collecting, segregating and recording waste 2.13 Cleaning the loom and attending routine lubrication points 2.14 Recording production, efficiency, stoppage and quality data
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Shuttle loom (power loom) 4.3 Rapier loom 4.4 Air-jet loom 4.5 Dobby and jacquard shedding attachment 4.6 Weft packages, pirns and weft accumulators 4.7 Machine setting sheets, design sheets and SOPs 4.8 GSM cutter and electronic balance 4.9 Tachometer and feeler gauges 4.10 Hand tool set and knotting tools 4.11 Oil can, grease gun and lubrication chart 4.12 Industrial vacuum cleaner and air blowing gun 4.13 Waste segregation bins 4.14 Learning manual and sample production formats 4.15 Multimedia projector, laptop and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared the loom for operation. 1.2 Carried out the weaving operation. 1.3 Attended warp and weft breakages and loom stoppages. 1.4 Monitored fabric quality and controlled loom faults. 1.5 Handled waste, cleaning and shift records.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning

	2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM CIRCULAR KNITTING OPERATIONS	Nominal Duration: 60 Hrs.	Unit Code: SICIP-TEX-PPFM-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform circular knitting operations. It specifically includes the tasks of preparing the circular knitting machine for operation, carrying out the circular knitting operation, attending yarn breakage, needle faults and machine stoppages, and monitoring knitted fabric quality and maintaining records.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the circular knitting machine for operation	1.1 Work order, <u>fabric specification</u> and machine setting sheet are obtained and understood. 1.2 Machine, creel and work area are checked for cleanliness and readiness. 1.3 Safety devices and guards are checked before start-up. 1.4 Needles, sinkers and knitting elements are checked and replaced where required. 1.5 Yarn packages are creeled and threaded through the <u>yarn feeding path</u> correctly. 1.6 <u>Machine settings</u> are verified against the machine setting sheet.
2. Carry out the circular knitting operation	2.1 <u>Knitting parameters</u> are set and maintained according to the fabric specification. 2.2 Stitch length (loop length) is checked and adjusted to achieve the specified GSM. 2.3 Yarn positive feed and yarn tension are maintained uniformly across all feeders. 2.4 Fabric take-down tension and roll formation are maintained correctly. 2.5 Knit structure is produced according to the design and cam arrangement.
3. Attend yarn breakage, needle faults and machine stoppages	3.1 Yarn breakage is attended and yarn is re-threaded promptly using the standard method. 3.2 Broken, bent and damaged needles and knitting elements are identified and replaced safely. 3.3 <u>Machine stop motions</u> are attended and reset correctly. 3.4 Elastane and plating yarn feeding faults are attended where applicable. 3.5 Fabric holes and press-off are prevented and rectified within the operator's scope.

Elements of Competency	Performance Criteria
	3.6 Faults beyond the operator's scope are reported to the supervisor or mechanic.
4. Monitor knitted fabric quality and maintain records	4.1 Knitted fabric is checked on the machine at specified intervals. 4.2 <u>Common knitted fabric defects</u> are identified and their causes are determined. 4.3 GSM, wales per inch, courses per inch and fabric width are measured and verified. 4.4 Yarn and fabric waste are collected and segregated as per workplace procedure. 4.5 Production and quality data are recorded in the <u>prescribed format</u>.

Range of Variables:

Variable	Range (May include but not limited to)
1. Fabric specification	1.1 S/J, Rib, and Interlock 1.2 Machine gauge, cylinder diameter and number of feeders 1.3 Stitch length/loop length (quality adjustment parameter) 1.4 GSM (grey and finished) 1.5 Grey width, tube width and finished width 1.6 Wales per inch (WPI) and courses per inch (CPI)
2. Yarn feeding path	2.1 Creel and package holder 2.2 Yarn guide and ceramic eyelet 2.3 Yarn tensioner and side tensioner 2.4 Yarn detector and stop motion 2.5 Positive feed device (storage feeder, MPF, belt and pulley) 2.6 Yarn feeder and guide finger 2.7 Waxing device 2.8 Elastane feeder and elastane package holder
3. Machine settings	3.1 Machine gauge and cylinder diameter 3.2 Number of feeders in use 3.3 Cam arrangement (knit, tuck and miss) and cam track sequence 3.4 Machine speed (RPM) 3.5 Air blower pressure and lint extraction setting
4. Knitting parameters	4.1 Stitch length (loop length) per feeder 4.2 Yarn input tension 4.3 GSM and VDQ pulley adjustment setting
5. Machine stop motions	5.1 Yarn detector and yarn breakage stop motion 5.2 Needle detector 5.3 Fabric hole detector

Variable	Range (May include but not limited to)
	5.4 Fabric take-down and roll-full stop 5.5 Elastane breakage detector 5.6 Creel and yarn-end sensor 5.7 Door and guard interlock 5.8 Overload, oil pressure and power failure trip
6. Common knitted fabric defects	6.1 Hole and needle line (needle mark) 6.2 Drop stitch and press-off (cloth fall-out) 6.3 Barre, streakiness and count/lot mix 6.4 Sinker mark and sinker line 6.5 Oil line, oil spot and grease stain 6.6 Tuck and float defects and wrong structure 6.7 Uneven GSM and stitch length variation 6.8 Elastane missing, elastane breakage and elastane pull-out 6.9 Contamination and foreign fibre 6.10 Running stitch (ladder) and snarl 6.11 Spirality 6.12 Dirt mark, lint mark and fly contamination 6.13 Wrong knit structure, wrong pattern and wrong colour sequence 6.14 Crease mark and roll edge damage
7. Prescribed format	7.1 Knitting production and efficiency report 7.2 GSM and stitch length record 7.3 Machine stoppage and breakdown register 7.4 Needle and sinker replacement record 7.5 Fabric roll card 7.6 On-machine inspection sheet 7.7 Waste record 7.8 Cleaning and lubrication checklist 7.9 Shift handover log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Classification of knitting machines and the working principle of weft knitting. 1.2 Construction of a circular knitting machine and the function of each part. 1.3 Latch needle, sinker and cam action in loop formation. 1.4 Knit structures, cam arrangements and needle selection for each structure. 1.5 Positive feed systems and their importance in maintaining uniform loop length. 1.6 Yarn requirements for knitting, waxing and yarn conditioning.
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	1.7 Knitting parameters and their effect on fabric quality and productivity. 1.8 Machine stop motions and their correct resetting. 1.9 Common knitted fabric defects, their causes and remedies. 1.10 Waste generation in knitting and permissible waste limits. 1.11 Safe working procedures, machine guarding and lockout-tagout in the knitting floor. 1.12 Recording and reporting production, stoppage and quality data.
2. Underpinning Skills	2.1 Checking the machine, creel, safety devices and guards before start-up 2.2 Creeling yarn packages and threading the yarn feeding path 2.3 Starting and stopping the machine following SOPs 2.4 Maintaining fabric take-down tension and roll formation 2.5 Producing the specified knit structure through correct cam arrangement 2.6 Attending yarn breakage and re-threading promptly 2.7 Attending and resetting machine stop motions 2.8 Preventing and rectifying holes and press-off 2.9 Identifying knitted fabric defects and determining their causes 2.10 Doffing, weighing, marking and handling fabric rolls 2.11 Collecting, segregating and recording waste 2.12 Cleaning and blowing the machine and attending lubrication points 2.13 Recording production, efficiency, stoppage and quality data
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Single jersey circular knitting machine 4.3 Rib and interlock circular knitting machine 4.4 Yarn packages of different counts and types, including elastane 4.5 Spare needles, sinkers, cams and needle removal tools 4.6 Measuring tape, steel rule and yarn tension meter 4.7 Tachometer and feeler gauges

	4.8 Hand tool set 4.9 Oil can, grease gun and lubrication chart 4.10 Industrial vacuum cleaner and air blowing gun 4.11 Waste segregation bins 4.12 Machine setting sheets, SOPs, learning manual and sample production formats 4.13 Multimedia projector, laptop and audio-visual clips
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared the circular knitting machine for operation. 1.2 Carried out the circular knitting operation. 1.3 Attended yarn breakage, needle faults and machine stoppages. 1.4 Monitored knitted fabric quality and maintained records.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM FLAT BED KNITTING OPERATIONS	Nominal Duration: 30 Hrs.	Unit Code: SICIP-TEX-PPFM-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform flat-bed knitting operations. It specifically includes the tasks of preparing the flat bed knitting machine for operation, carrying out flat bed knitting operations, and checking panel quality, handling waste and maintaining records.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the flat bed knitting machine for operation	1.1 Work order, panel specification and knitting program are obtained and understood. 1.2 Safety devices and guards are checked before start-up. 1.3 <u>Needles and knitting elements</u> of the front and back bed are checked and replaced where required. 1.4 Yarn packages are creeled and threaded through the <u>yarn feeding path</u> correctly.
2. Carry out flat bed knitting operations	2.1 Machine settings are adjusted according to yarn count and design. 2.2 Carriage traverse, cam setting and yarn carrier movement are monitored during operation. 2.3 Yarn breakage and needle faults are attended promptly and safely.
3. Check panel quality, handle waste and maintain records	3.1 <u>Common flat bed knitting defects</u> are identified and their causes are determined. 3.2 Corrective actions within the operator's scope are taken to remove defects. 3.3 Rejected and defective panels are separated, marked and reported. 3.4 Yarn and fabric waste are collected and segregated as per workplace procedure. 3.5 Machine, needle bed and work area are cleaned as per the cleaning schedule. 3.6 Production and quality data are recorded in the <u>prescribed format</u> .

Range of Variables:

Variable	Range (May include but not limited to)
1. Needles and knitting elements	1.1 Latch needles of the front and back beds 1.2 Needle beds and needle tricks 1.3 Cams, cam boxes and cam settings 1.4 Carriage and carriage drive 1.5 Yarn carriers and carrier rails
2. Yarn feeding path	2.1 Creel and package holder 2.2 Yarn guide and ceramic eyelet 2.3 Main tensioner and side tensioner 2.4 Yarn stop motion and yarn detector 2.5 Yarn feeder and carrier
3. Common flat-bed knitting defects	3.1 Hole and dropped stitch 3.2 Needle line and needle mark 3.3 Short, oversized and out-of-tolerance panel 3.4 Press-off and fabric drop 3.5 Yarn tension mark and tight/loose course 3.6 Oil stain, dirt mark and lint contamination 3.7 Shade variation and lot mix
4. Prescribed format	4.1 Flat knitting production report 4.2 Panel measurement and weight record 4.3 Machine stoppage and breakdown register 4.4 Needle replacement record 4.5 Bundle and lot card 4.6 Quality checking and rejection sheet 4.7 Waste record 4.8 Shift handover log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Working principle and construction of the flat bed knitting machine. 1.2 Types of flat bed knitting machines 1.3 Machine gauge and its relationship with yarn count and panel appearance. 1.4 Knitting program, pattern chain, punch card and computerized design file. 1.5 Stitch density, take-down tension and their effect on panel dimensions. 1.6 Common flat bed knitting defects, their causes and remedies. 1.7 Needle changing procedure and precautions. 1.8 Production and efficiency calculation for flat bed knitting machines.
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	1.9 Waste generation in flat bed knitting and permissible waste limits. 1.10 Safe working procedures and machine guarding on the flat knitting floor.
2. Underpinning Skills	2.1 Checking the machine, creel, safety devices and guards before start-up 2.2 Checking and replacing needles and knitting elements of both beds safely 2.3 Adjusting machine settings according to yarn count and design 2.4 Maintaining uniform stitch density and yarn tension 2.5 Performing racking, transfer and shaping operations 2.6 Attending yarn breakage and needle faults promptly 2.7 Producing tipping, stripes and structure changes correctly 2.8 Separating, counting and bundling panels 2.9 Attaching shade, lot and size identification to bundles 2.10 Identifying flat-bed knitting defects and determining their causes 2.11 Collecting, segregating and recording waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Flat-bed knitting machine (manual/semi-automatic/computerized) 4.3 Creel with tensioners, guides and stop motions 4.4 Yarn packages of different counts, types and shades 4.5 Spare needles, transfer points and needle removal tools 4.6 Electronic balance and counting glass 4.7 Oil can and lubrication chart 4.8 Panel samples showing common flat bed knitting defects 4.9 Bundling and identification materials 4.10 Waste segregation bins 4.11 Knitting programs, SOPs, learning manual and sample production formats 4.12 Multimedia projector, laptop and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared the flat bed knitting machine for operation. 1.2 Carried out flat bed knitting operations. 1.3 Checked panel quality, handled waste and maintained records.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM FABRIC INSPECTION, GRADING AND DEFECT IDENTIFICATION	Nominal Duration: 40 Hrs.	Unit Code: SICIP-TEX-PPFM-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform fabric inspection, grading and defect identification. It specifically includes the tasks of preparing for fabric inspection, carrying out fabric inspection, identifying fabric defects and determining their causes, applying the fabric grading system, and reporting inspection results and taking corrective action.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for fabric inspection	1.1 Work order, <u>fabric specification</u> and buyer requirements are obtained and understood. 1.2 Inspection machine, lighting arrangement and work area are checked for readiness. 1.3 Inspection tools and instruments are collected and checked for accuracy.
2. Carry out fabric inspection	2.1 Fabric is inspected on the <u>face side</u> at the specified speed and viewing distance. 2.2 Fabric parameters are measured and verified against the specification. 2.3 Defects are marked with stickers, tags or flags at the selvedge.
3. Identify fabric defects and determine their causes	3.1 Woven fabric defects are identified and classified. 3.2 Knitted fabric defects are identified and classified. 3.3 Major and minor defects are distinguished according to the standard. 3.4 <u>Repairable defects</u> are separated for mending.
4. Apply the fabric grading system	4.1 <u>Fabric grading system</u> is applied as per the buyer or organizational requirement. 4.2 Penalty points are assigned to defects according to defect length and severity. 4.3 Points per 100 square yards or square metres are calculated correctly. 4.4 Fabric is graded as acceptable or rejected against the standard limit. 4.5 Grade and inspection results are recorded roll-wise.
5. Report inspection results and take corrective action	5.1 Inspection results are recorded in the <u>prescribed format</u> . 5.2 Inspection reports are prepared and submitted to the authority.

Elements of Competency	Performance Criteria
	5.3 Mending and spot cleaning are arranged for repairable defects.

Range of Variables:

Variable	Range (May include but not limited to)
1. Fabric specification	1.1 Fabric construction (EPI x PPI for woven, WPI x CPI for knitted) 1.2 Weave or knit structure 1.3 GSM
2. Face side	2.1 Right side (technical face) of woven fabric 2.2 Technical face of knitted fabric 2.3 Identification of face by weave or knit appearance 2.4 Selvedge marking and pin mark indication 2.5 Direction of fabric (warp direction and course direction)
3. Repairable defects	3.1 Loose thread and floating yarn 3.2 Minor slub and protruding knot 3.3 Small drop stitch 3.4 Oil and dirt spot removable by spot cleaning 3.5 Missing end that can be lashed-in 3.6 Minor selvedge fault 3.7 Loose weft end at the selvedge
4. Fabric grading system	4.1 4-point system (most widely used for apparel fabric) 4.2 10-point system 4.3 Grade A, B and reject classification
5. Prescribed format	5.1 Fabric inspection report 5.2 4-point grading sheet 5.3 Roll-wise summary sheet 5.4 Shade grouping and shade band record 5.5 Defect analysis sheet 5.6 Rejection and second-quality record 5.7 Mending record 5.8 Delivery and handover challan

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Fabric specification, buyer requirements and tolerance limits. 1.2 Construction and operation of the fabric inspection machine. 1.3 Identification of the face side and direction of woven and knitted fabric.
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	1.4 Measurement of fabric width, length, weight, GSM, EPI, PPI, WPI and CPI. 1.5 Skewness, bowing and spirality: causes and measurement. 1.6 Woven fabric defects, their appearance, causes and responsible process. 1.7 Knitted fabric defects, their appearance, causes and responsible process. 1.8 Fabric grading systems (4-point, 10-point, 2.5-point and Graniteville '78'). 1.9 Calculation of penalty points per 100 square yards and per 100 square metres. 1.10 Documentation, reporting and traceability requirements.
2. Underpinning Skills	2.1 Interpreting fabric specifications and buyer quality requirements 2.2 Checking the inspection machine, lighting and instruments for readiness 2.3 Identifying the face side and direction of the fabric 2.4 Inspecting fabric at the specified speed and viewing distance 2.5 Counting EPI, PPI, WPI and CPI using a pick glass 2.6 Checking shade variation against the approved swatch and shade band 2.7 Identifying and classifying woven fabric defects 2.8 Identifying and classifying knitted fabric defects 2.9 Applying the 4-point and other grading systems 2.10 Segregating and identifying rejected and second-quality fabric 2.11 Handing over inspected fabric with correct documentation
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest in duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Fabric inspection machine with light box 4.3 Colour matching cabinet with multiple light sources 4.4 Fabric samples showing a wide range of defects 4.5 GSM cutter, GSM balance and electronic balance 4.6 Measuring tape, steel rule and length counter 4.7 Fabric trolley and roll-handling arrangement

	4.8 Inspection formats, grading sheets and buyer quality manuals 4.9 Learning manual, SOPs and reference standards 4.10 Multimedia projector, laptop and audio-visual clips
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared for fabric inspection. 1.2 Carried out fabric inspection. 1.3 Identified fabric defects and determined their causes. 1.4 Applied the fabric grading system. 1.5 Reported inspection results and took corrective action.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

WORKSHOP / LAB FACILITY STANDARD

Course Name:	Production Process in Fabric Manufacturing
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

1. Classroom - 400 sft (37 sqm)
2. Workshop/Lab - 1800 sft (167 sqm), **OR**
3. Classroom cum workshop - 2100 sft (195 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training Facilities	Required Quantity
1.	Computer/Laptop	1
2.	Multimedia projector & screen	1
3.	Printer	1
4.	Scanner	1
5.	Internet connection with Wi-Fi facility	1
6.	Whiteboard with accessories	1
7.	Cone/cheese winding machine (drum winder)	1
8.	Pirn winding machine	1
9.	Direct (beam) warping machine with creel	1
10.	Sectional warping machine with creel	1
11.	Sizing machine with drying arrangement	1
12.	Size cooking vessel with stirrer and service tank	1 set
13.	Drawing-in frame with stand and accessories	2 sets
14.	Healds, heald frames, reeds and drop wires (assorted)	As required
15.	Warp tying/knotting machine	1
16.	Shuttle loom (power loom)	1
17.	Rapier loom	1
18.	Air-jet loom	1
19.	Dobby shedding attachment/dobby loom	1
20.	Jacquard shedding attachment/jacquard loom	1

S.N.	Major Equipment and Training Facilities	Required Quantity
21.	Single jersey circular knitting machine	1
22.	Rib/interlock circular knitting machine	1
23.	Flat bed knitting machine (manual/semi-automatic)	2
24.	Computerized flat bed knitting machine	1
25.	Fabric inspection machine with light box	1
26.	Colour matching cabinet (D65, TL84, CWF, UV)	1
27.	Fabric folding and rolling table with length counter	1
28.	GSM cutter and GSM balance	2 sets
29.	Electronic balance (0.001 g and 0.01 g accuracy)	2
30.	Pick glass/counting glass and magnifier	10
31.	Stitch length (course length) measuring device	5
32.	Yarn tension meter	2
33.	Reed hook, drawing hook and denting hook set	10 sets
34.	Needle, sinker and needle removal tool set	5 sets
35.	Hand tool set (spanner, screwdriver, plier, nipper)	5 sets
36.	Measuring tape and steel rule	25
37.	Viscometer/Brix meter, pH meter and thermometer	1 set
38.	Material handling trolley for beams, packages and fabric rolls	4
39.	Industrial vacuum cleaner/air blowing gun	2
40.	Lockout-tagout kit	2 sets
41.	Waste segregation bins (colour coded)	4
42.	Tool board and storage cabinet	2
43.	Fire extinguisher	2
44.	First aid box	1
45.	PPE set (dust mask, ear plug, gloves, apron, cap, safety shoes)	25 sets

The following conditions must be fulfilled –

1. The institute shall not use the same facilities for any other projects/organizations offering a similar course.
2. The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facilities required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirements of the Competency Standard (CS) provided by SICIP.

For the operation of the training course on Production Process in Fabric Manufacturing, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

Where a training institute does not possess the full-scale production machinery, the institute must establish a formal linkage with a weaving and/or knitting factory so that the practical sessions of the occupation-specific units can be conducted in an actual workplace under supervision.

The institute will also provide all other hand tools and power tools as per the CS for 25 trainees. The institute will also arrange adequate seating arrangements and classroom setup for the 25 trainees.